

Get Started with Docker



COMS4156 Demo, Fall 2023

Docker Quickstart

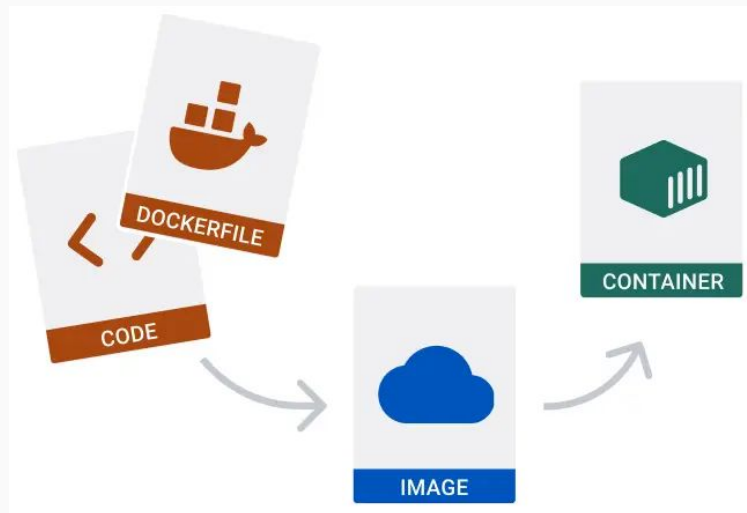
1. What is Docker?
2. Why should I use it?
3. How does it work?

What is docker?

- Image and Containers

Main ingredients:

1. Your project code
2. Dockerfile



What is docker?

- Image and Containers

What does docker do?

1. Create your own 'OS'.

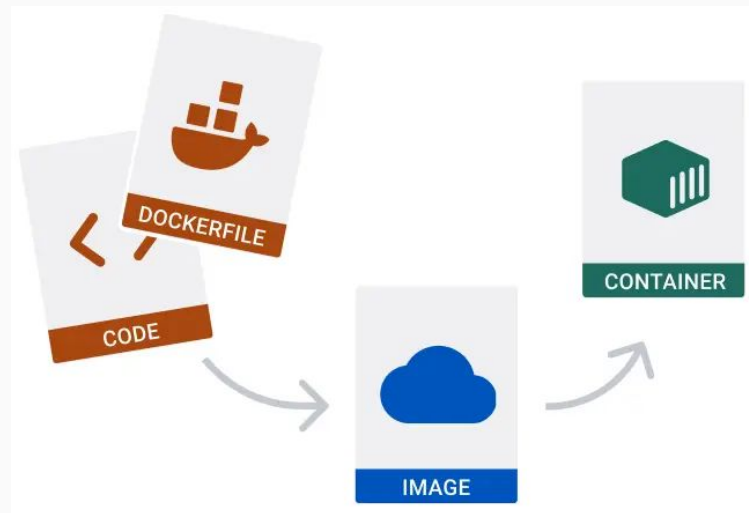
Input: code + Dockerfile

Output: an Image

2. Run (lightweight) your OS

Input: an Image

Output: containers

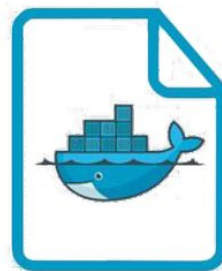


What is docker?

- Image and Containers

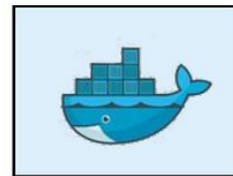
Image:

- An image is a read-only template with instructions *for creating a Docker container*.



Dockerfile

Build



Docker
Image

What is docker?

- Image and Containers

Image:

- An image is a read-only template with instructions *for creating a Docker container*.
- Often, an image is *based on another image*, with some additional customization.

```
FROM node:12.18.1
```

```
WORKDIR /app
```

```
### lazy version: just copy over everything from current location  
COPY . .
```

```
### full version: just copy over the necessary source code  
# COPY package.json package.json  
# COPY package-lock.json package-lock.json  
# RUN npm install  
# COPY server.js
```

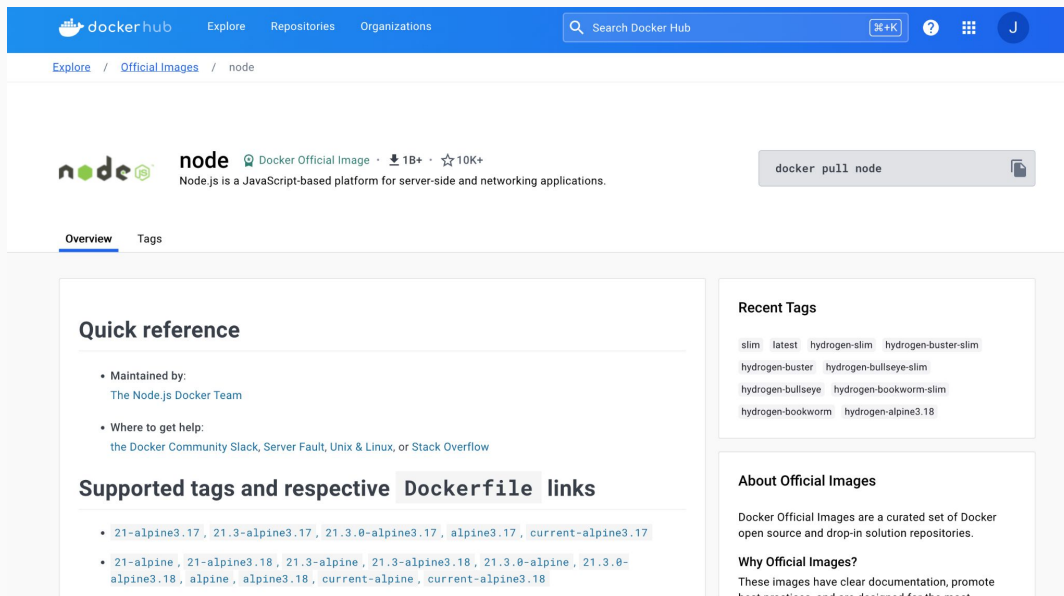
```
CMD [ "node", "server.js" ]
```

An example dockerfile

What is docker?

- Image and Containers

Many images are already built for all kinds of applications!



The screenshot shows the Docker Hub interface for the 'node' image. The top navigation bar is blue with the Docker Hub logo, 'Explore', 'Repositories', and 'Organizations' links. A search bar is on the right. Below the navigation bar, the breadcrumb trail reads 'Explore / Official Images / node'. The main content area features the 'node' logo, the text 'node Docker Official Image · 1B+ · 10K+', and a description: 'Node.js is a JavaScript-based platform for server-side and networking applications.' A 'docker pull node' button is on the right. Below this, there are tabs for 'Overview' and 'Tags'. The 'Overview' tab is active, showing a 'Quick reference' section with links for 'Maintained by' (The Node.js Docker Team) and 'Where to get help' (the Docker Community Slack, Server Fault, Unix & Linux, or Stack Overflow). Below this is a section titled 'Supported tags and respective Dockerfile links' with two bullet points listing various tags like '21-alpine3.17', '21.3-alpine3.17', '21.3.0-alpine3.17', 'alpine3.17', 'current-alpine3.17', '21-alpine', '21-alpine3.18', '21.3-alpine', '21.3-alpine3.18', '21.3.0-alpine', '21.3.0-alpine3.18', 'alpine', 'alpine3.18', 'current-alpine', and 'current-alpine3.18'. On the right side, there are two sections: 'Recent Tags' with a list of tags like 'slim', 'latest', 'hydrogen-slim', 'hydrogen-buster-slim', 'hydrogen-buster', 'hydrogen-bullseye-slim', 'hydrogen-bullseye', 'hydrogen-bookworm-slim', 'hydrogen-bookworm', and 'hydrogen-alpine3.18'; and 'About Official Images' with a description of Docker Official Images and a section titled 'Why Official Images?'.

node Docker Official Image · 1B+ · 10K+
Node.js is a JavaScript-based platform for server-side and networking applications.

docker pull node

Overview Tags

Quick reference

- Maintained by:
The Node.js Docker Team
- Where to get help:
the Docker Community Slack, Server Fault, Unix & Linux, or Stack Overflow

Supported tags and respective Dockerfile links

- 21-alpine3.17, 21.3-alpine3.17, 21.3.0-alpine3.17, alpine3.17, current-alpine3.17
- 21-alpine, 21-alpine3.18, 21.3-alpine, 21.3-alpine3.18, 21.3.0-alpine, 21.3.0-alpine3.18, alpine, alpine3.18, current-alpine, current-alpine3.18

Recent Tags

slim latest hydrogen-slim hydrogen-buster-slim
hydrogen-buster hydrogen-bullseye-slim
hydrogen-bullseye hydrogen-bookworm-slim
hydrogen-bookworm hydrogen-alpine3.18

About Official Images

Docker Official Images are a curated set of Docker open source and drop-in solution repositories.

Why Official Images?

These images have clear documentation, promote best practices, and are designed for the most

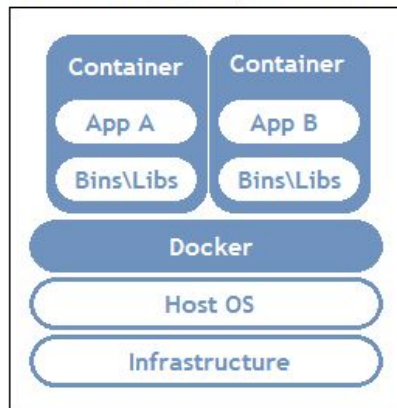
What is docker?

- Image and Containers

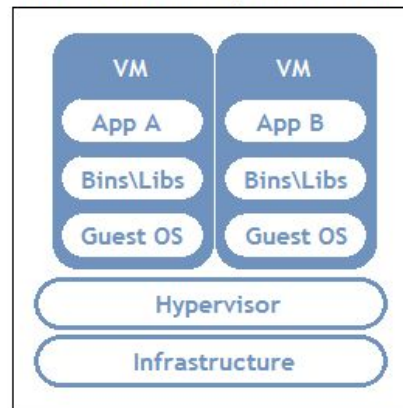
Container:

- A container is a runnable *instance* of an image.

Container Based Implementation



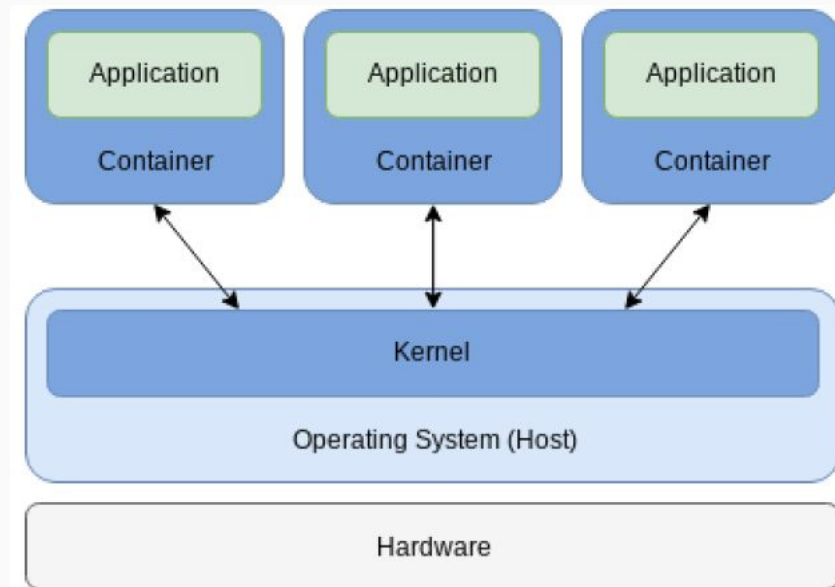
Virtual Machine Implementation



Why should I use it?

- Benefits

- Isolated environment
- Portability
- Scalability

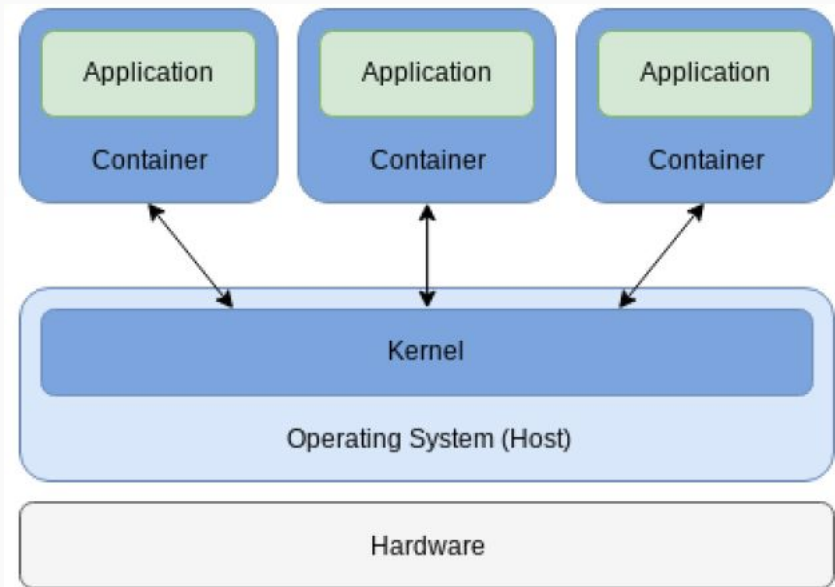


Why should I use it?

- Benefits

From my experience:

- Huge relief on software dependency issues
- Can **sudo** inside the container

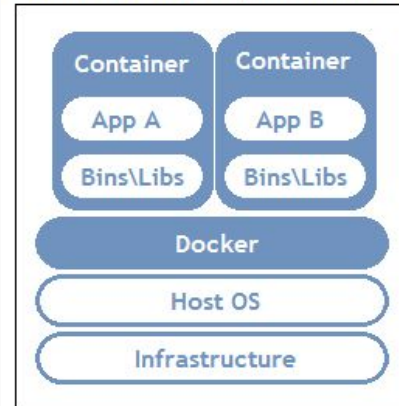


Why should I use it?

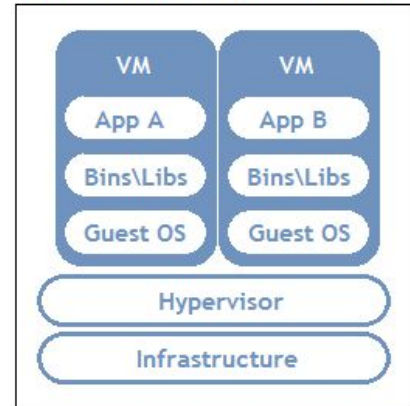
- Drawbacks

- Sometimes need a lot of configurations to get it right

Container Based Implementation



Virtual Machine Implementation



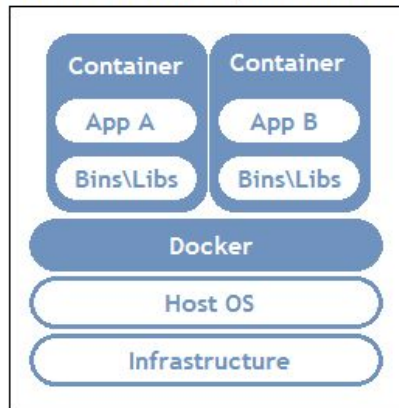
Why should I use it?

- Drawbacks

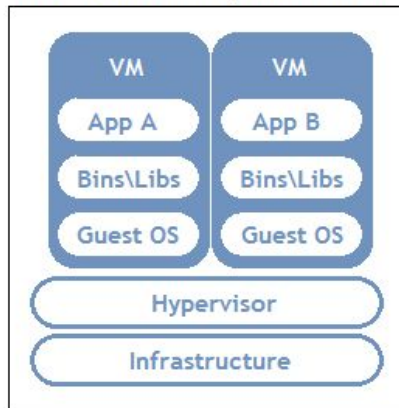
From my experience:

- Cannot adjust container configurations after launch
- An image is huge in disk size

Container Based Implementation



Virtual Machine Implementation



How does it work?

- Demo

Github link: [jasonyux/COMS4156_demo \(github.com\)](https://github.com/jasonyux/COMS4156_demo)